

NAKAYAMA STEEL WORKS, LTD. (5408 JP) — Equity Research Deep-Dive

5408 JP Steel · TSE Prime Osaka, Japan · FY-end 31 March · Report date 31 July 2026 · Currency ¥ (JPY)

Sector: Steel / Iron & Steel (Non-energy minerals) · **Exchange:** Tokyo Stock Exchange, Prime Market · **Ticker:** 5408 JP / 5408.T **Fiscal year-end:** 31 March · **HQ:** Osaka, Japan · **Founded:** 1919 (incorporated 1923; 中山製鋼所 from 1934) **Report date:** 31 July 2026 · **Reporting currency:** Japanese Yen (¥)

Rating framing: This note is a research dossier, not a formal buy/sell call. Where a house view is expressed it is labelled as such.

DATA-VINTAGE & SOURCING NOTE (read first). Nakayama is a Japanese small-cap; granular data is dispersed across Japanese-language IR/aggregator sources (the issuer's own domain and several aggregators were not directly retrievable from this environment, so figures were triangulated across public search results — Nikkei, Kabutan, IR Bank, MarketScreener, Yahoo! Finance Japan, StockAnalysis, Japan Metal Daily, MONOist, company disclosures). Headline figures (net sales, ordinary profit, guidance, the Nippon Steel EAF JV, share price/valuation, board/leadership change) are corroborated. **Second-decimal precision on some line items — particularly segment operating profit, multi-year EBITDA, and peer multiples — should be treated as indicative and verified against the 決算短信 (Tanshin), 有価証券報告書 (Yūkashōken Hōkokusho / annual securities report) and the FY3/26 統合報告書 (Integrated Report) before use in a live model.** Cells that are estimates carry "(e)".

0. Executive Summary / Investment Case in Brief

What it is. Nakayama Steel Works is a **sub-scale Osaka electric-arc-furnace (EAF) steelmaker** — the ~4th-largest Japanese ordinary-steel EAF producer, unusually flat-product-tilted (hot-rolled coil, thick/medium plate, checkered plate) alongside long products (bar, wire rod, rebar). It shut its blast furnace in 2002, survived a 2013 state-backed (REVIC) rescue, and today melts scrap + purchased slab in a ~600kt/yr EAF and re-rolls it, with tiny Engineering (marine reefs, rolls, valves) and Real Estate (Osaka Bay land) segments (~2% of sales combined).

The one-line thesis. A **deep-value, net-cash, 0.31x-book** cyclical with a **transformational growth option** — the **NN Steel LLC joint venture with Nippon Steel (Nakayama 51% / Nippon Steel 49%, formed 1 Apr 2026)** to build a ~1.2Mt low-CO₂ EAF (~2× capacity, ~¥95–105bn, start-up ~2030) that internalises slab and gives Nippon Steel an offtake — fighting a **weak, import-pressured steel spread**, a **four-year earnings decline**, and a **heavy capex cycle**. It is best understood as a **cheap, asset-backed call option on Nippon Steel's Kansai EAF/green-steel strategy**, not an earnings-multiple stock.

Snapshot (as-of late Jul 2026)	Value
Price / Market cap	~¥634 / ~¥34bn (¥34–38bn incl. treasury)
P/B (PBR)	~0.31x (BPS ¥2,013; near decade-low)
EV/EBITDA / EV/Sales	~4.1x / ~0.22x
P/E (ttm / fwd)	~14x / ~8–10x
Net debt/EBITDA	~0.3x (net cash ~¥6bn; 72% equity ratio)
Dividend yield	~2.2% (¥14) → ~3.3% (¥20e FY3/27)
FY3/26 sales / ordinary / net income	¥148.3bn (~12.4%) / ¥4.81bn (~40.8%) / ¥2.46bn (~56.8%)
FY3/27 guide (ordinary / net income / DPS)	¥2.0bn (~58.4%) / ¥3.5bn (+42%, special gains) / ¥20
ROE (FY3/26 / target)	2.28% / ~7%
Top holder / Nippon Steel stake	Hanwa 14.87% / Nippon Steel 0% (exited Dec-2022)

Bull in one line: 0.31x book with net cash and latent Osaka land = a hard floor; the Nippon Steel JV permanently removes the slab-purchase drag, adds green-steel/GX-subsidy upside, and makes Nakayama a clean consolidation/take-out candidate at 0.7–1.0x book (~¥1,400–2,000/share). **Bear in one line:** a sub-scale, price-taking commodity EAF with ~5% EBITDA margins and structurally high Kansai power costs can earn below its cost of capital *forever* — 0.31x book may be the *correct* price for a value-destroyer about to sink ¥100bn into an oversupplied product; the "dividend hike" is one-off-land-gain-funded and not a sustainable run-rate. **The crux:** the tension between a **0.3x-book, net-cash floor** and a **collapsing, sub-cost-of-capital earnings trajectory** (FY3/27 ordinary just ¥2.0bn). Re-rating needs a catalyst — ROE inflection, capital-return escalation, JV proof-points, or a Nippon Steel move — none of which is guaranteed, and any of which could take years. **Skew is positive; the risk is "dead money," not a zero.**

(Full bull/bear, pre-mortem, contrarian view — Section 19; scenario valuation — Section 20; forensic — Section 21; short-seller — Section 22.)

1. Layman's Guide: What the Company Does & How It Makes Money

In one paragraph. Nakayama Steel Works melts recycled steel scrap (and re-melts/re-rolls purchased semi-finished steel "slabs") in an **electric arc furnace** at its Funamachi works on Osaka Bay, then rolls that steel into flat products (hot-rolled coil/sheet, thick and medium **plate**, checkered/patterned "非常口" floor plate) and **long products** (steel bars, wire rod, bar-in-coil, rebar). It sells these mostly to Japanese trading houses and processors serving construction, civil engineering, shipbuilding and machinery. Around this core steel business it runs two much smaller units: an **Engineering** unit (steel marine structures — artificial fish reefs and aquaculture reefs — plus rolls and industrial valves) and a **Real Estate** unit that leases and develops land, much of it former plant land on the Osaka waterfront.

Division	What it sells	How it earns	Rough scale
Iron & Steel	HR coil/sheet, plate, checkered plate, bars, wire rod, rebar, slab, by-products (slag)	Spread: product price – (scrap + purchased slab + electricity + labour) × volume	~90%+ of sales; the swing factor for group profit
Engineering	Steel fish reefs / aquaculture reefs, rolls, valves, marine & civil steel structures, construction	Project/order-based margins; counter-cyclical, public-works linked	Small (low-to-mid single-digit % of sales)
Real Estate	Leasing & sale of owned land/buildings (ex-plant Osaka Bay land)	Rental yield; occasional development/disposal gains	Small by sales but structurally high-margin ; a hidden asset-value story

The one-sentence "how it makes money": it is a **spread business** — group profit is driven far more by the gap between steel selling prices and input costs (scrap, purchased slab, electricity) than by revenue growth, which is why an 8–12% fall in sales can drop ordinary profit by 30–58%. Real Estate provides a steady, high-margin cash cushion and an under-appreciated NAV; Engineering is a small, lumpy, public-works-linked diversifier.

2. Business Model & Key Drivers

Business model in simple terms. Nakayama is a **mini-mill / EAF re-roller** sitting in the mid-tier of the Japanese steel hierarchy — far smaller than the blast-furnace majors (Nippon Steel, JFE, Kobe Steel) but with a **flat-product (plate/coil) tilt unusual for an EAF player**, most of which make long products (rebar/bar). It shut its **blast furnace and converter in 2002** and has since run an EAF-plus-rolling model, historically **buying semi-finished slab** (including from Nippon Steel) to feed rolling mills whose capacity exceeds its own steelmaking. The Nippon Steel JV new EAF is designed precisely to close that gap — internalising slab supply and turning Nakayama into a net slab producer.

Key business drivers (ranked by profit sensitivity):

- Steel spread (metal margin)** — selling price of plate/coil/bar minus scrap and purchased-slab cost. The single largest driver; compressed in FY3/25–26 by weak domestic demand and cheap Chinese/Asian imports.
- Scrap price** — principal raw material for the EAF (ferrous scrap). Volatile; a pass-through lag hurts margins when scrap rises faster than product prices.
- Electricity cost** — EAF steelmaking is power-intensive; Japanese industrial power prices (and fuel-cost adjustments) are a major, partly uncontrollable cost line. A structural EAF disadvantage vs. cheaper-power geographies.
- Volume / utilisation** — crude-steel output and mill utilisation. Existing EAF target is a **stable ~50,000 t/month (~600kt/yr)**; the **No. 5 substation accident (26 Sep 2025) that halted the EAF for ~3 months into 24 Dec 2025 (~¥1.6bn hit, FY3/26 H2)** shows how sensitive profit is to unplanned single-site downtime.
- Product mix** — higher-value plate and specialty vs. commodity rebar/bar. Mix is a lever Nakayama can partially control.
- Construction & civil-engineering demand** (domestic) — the dominant end-market; sensitive to Japanese public works, non-residential construction and the construction labour shortage.
- Import competition & FX** — low-priced imports (esp. China) cap domestic prices; a weak yen helps exports/import-substitution but raises imported input and energy costs.
- Capex cycle / decarbonisation** — the ¥95–105bn JV EAF is a multi-year capex supercycle for a company this size, with subsidy support (GX / green-transformation) and Nippon Steel co-investment shaping returns.

Contracts & payment terms (typical for the sub-sector; company-specific terms not fully public): Japanese steel is sold substantially through **trading houses (商社)** on a mix of **"himo-tsuki"** (tied, contract) volumes and spot; pricing on many flat products is negotiated quarterly/periodically with scrap-linked formulas; long products track published market/scrap indices. Standard domestic B2B terms (invoice + promissory-note/bank transfer, ~end-of-month-plus-30/60 settlement) apply. The Nippon Steel JV adds a **long-term slab-offtake arrangement** (Nippon Steel purchasing part of the JV EAF's slab), which will over time convert a portion of Nakayama's output into contracted, captive volume.

3. Macro & Micro Factors — Tailwinds vs. Headwinds

Tailwinds - Decarbonisation / green-steel premium. EAF steel is far lower-CO₂ than blast-furnace steel. Nakayama's all-EAF route and the new low-CO₂ JV EAF position it for green-steel demand and **GX (green transformation) subsidies**; company targets a **46% supply-chain CO₂ cut by FY2030 vs. FY2013** (≈80% of its footprint is Scope 3 — i.e., purchased iron units, which the new EAF reduces). - **Nippon Steel alliance & JV.** Deep-pocketed strategic partner co-investing ~¥50bn, providing technology, slab-offtake and de-risking the largest capex in Nakayama's history. - **Japan structural EAF shift.** Japanese majors (Nippon Steel, JFE, Kobe) are themselves pivoting blast furnaces to EAFs; Nakayama is an early-mover incumbent EAF operator with a flat-product niche. - **Weak yen** aids import-substitution economics for domestic mills and any export volumes. - **TSE low-PBR reform** pressure to improve capital efficiency and shareholder returns at sub-1x-book names — Nakayama trades ~0.3x book. - **Real-estate NAV.** Osaka Bay land carried at low historical cost; latent value and redevelopment optionality (Osaka growth: Expo 2025 legacy, IR/casino, urban redevelopment).

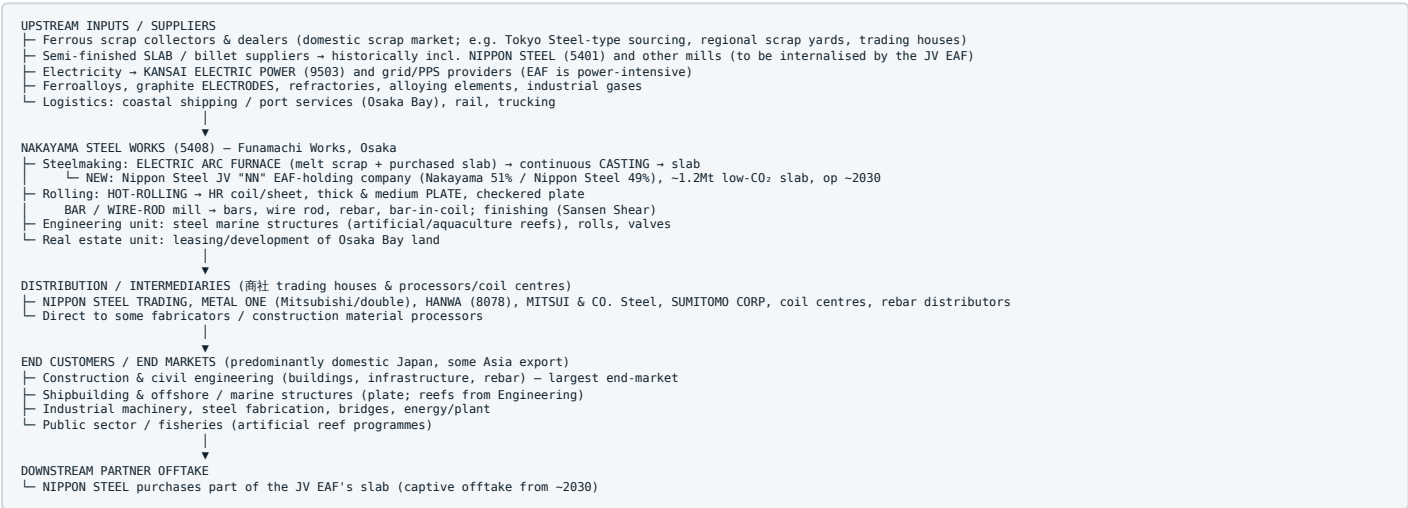
Headwinds - Cheap imports. Chinese/Asian export flood depresses Japanese domestic steel prices and spreads — the dominant near-term margin pressure. - **Weak domestic demand.** Sluggish construction (labour shortages, "2024 problem" logistics/overtime caps delaying projects), soft manufacturing. - **Energy costs.** Elevated Japanese electricity prices structurally penalise EAF economics. - **Scale disadvantage.** Sub-scale vs. majors and vs. large EAF peers (Tokyo Steel); a price-taker on commodity grades. - **Capex & execution risk.** ¥95–105bn JV EAF (through ~2030) strains a ¥33bn-market-cap balance sheet; cost overruns, delayed ramp, or a weak steel market at start-up would pressure returns and free cash flow. - **Cyclicality & single-site concentration.** Heavy reliance on one Osaka works; the **Sep–Dec 2025 (FY3/26) substation accident** illustrates single-point-of-failure risk. - **Four consecutive years of profit decline** into FY3/26 and a further guided drop in FY3/27 — earnings momentum is firmly negative.

Perceived value / branding. Steel is largely a commodity; Nakayama is not a premium brand and competes primarily on price, reliability, specification and logistics rather than brand. Its genuine differentiators are (i) a flat-product (plate & checkered plate) franchise rare among EAF makers, (ii) Osaka Bay location with deep-water/coastal logistics serving the Kansai industrial belt, and (iii) the Nippon Steel technical and supply relationship.

Economic moats — assessment: Narrow-to-none, with two partial, durable edges. - Cost/location advantage (partial): integrated coastal site, established rolling assets, proximity to Kansai customers and scrap supply. - Relationship moat (partial but real): the Nippon Steel capital/business alliance and now the 51/49 EAF JV create a quasi-captive, hard-to-replicate partnership — technology transfer, slab offtake and co-funding that a standalone sub-scale mill could not obtain. This is the closest thing to a moat. - Low-CO2 / regulatory positioning (emerging): first-mover EAF status could become a moat if green-steel procurement and carbon pricing reward it — but this is an option, not yet a cash moat. - What it lacks: scale economies, proprietary product, switching costs, or pricing power on commodity grades. On plain vanilla product it is a price-taker.

Bargaining power: - vs. customers (trading houses / processors): Weak-to-moderate. Sales are intermediated by powerful trading houses (Nippon Steel Trading, Metal One, Hanwa, Mitsui & Co. Steel, etc.); on commodity grades customers hold the whip hand, though the plate niche and JIT logistics give some leverage. - vs. suppliers (scrap dealers, electricity, slab): Weak. Scrap is a market-priced input; electricity is effectively administered/oligopolistic (Kansai Electric); purchased slab historically dependent on partners. The JV EAF improves supplier power over time by internalising slab. - vs. the strategic partner (Nippon Steel): Asymmetric. Nippon Steel is technology provider, JV co-owner (49% of NN Steel LLC), slab supplier and future slab/coil offtaker — a source of strength (de-risking) but also of dependence and related-party risk on JV terms and offtake pricing. Note: Nippon Steel is no longer an equity holder in Nakayama itself — it fully exited the share register in December 2022 (it had entered as a 2013 rehabilitation sponsor). The relationship is now contractual (alliance + JV), not ownership. - vs. the largest shareholder (Hanwa, 8078): Notable related party. Hanwa is simultaneously Nakayama's top shareholder (~14.9%) and a principal distributor/off-taker of its steel — a dual role that aligns interests but concentrates channel power and warrants RPT scrutiny.

5. Supply Chain Map — Upstream Inputs to End Customer



Named counterparties above are the most-cited in public sources; a full customer/supplier roster is not disclosed. Company tickers shown where listed.

6. Latest Earnings Analysis — FY3/26 Full Year, Guidance & Interim Trend

Headline (FY ended 31 March 2026, consolidated; announced 8 May 2026):

Metric (¥bn)	FY3/25	FY3/26	YoY	vs guidance
Net sales	169.3	148.3	-12.4%	vs revised guide ¥151.0bn → slight miss
Operating income	~8.4	~4.2 (e)	~-50%	≈ in line (exact figure vs Tanshin)
Ordinary (recurring) profit	8.1	4.806	-40.8%	Beat revised guide ¥4.0bn by ~¥0.8bn
Net income (owners)	5.70	2.462	-56.8%	Beat revised guide ¥2.3bn
EPS (¥)	~105 (e)	~45.4 (e)	down	—
DPS (¥)	13 (e)	14	+¥1	—

- Beat/miss: ordinary profit fell 40.8% YoY but beat the company's own (already cut) forecast — Japanese wires (Fisco/iFIS) headlined "経常40.8%減益・事前予想を上回る." A "less-bad-than-feared" print, not a beat of a rising bar. This was the 4th straight year of profit decline (ordinary profit trajectory: ~¥12.2bn FY3/24 → ¥8.1bn FY3/25 → ¥4.8bn FY3/26).
- The decisive event — the No. 5 substation accident. On 26 Sep 2025 the No. 5 substation failed and halted the EAF for ~3 months; operation was restored 24 Dec 2025 (~¥1.6bn estimated hit). Downstream rolling kept running on higher-cost purchased iron sources. This drove a 31 Oct

2025 downward revision (FY net-income guide ¥4.2bn → ¥2.3bn) and explains the H2 collapse. (Note: this accident is an FY3/26 H2 event — earlier press summaries that pinned it to "FY2025" refer to the year ending March 2026.)

- **Interim shape:** H1 FY3/26 (Apr–Sep 2025): sales **¥76.6bn (–14.4%)**, operating income **¥3.02bn (–36.9%)**, ordinary **¥2.93bn (–37.6%)**, net income **¥1.71bn (–46.7%)**. Implied H2: ordinary ~¥1.87bn, net ~¥0.75bn — the accident quarter.
- **Why margins fell:** (1) weak domestic construction demand; (2) **low-priced imports** (China/Asia) compressing spreads; (3) the **substation accident** and higher-cost substitute iron; (4) elevated energy (Kansai Electric 2024 tariff hikes), material and labour costs.

Guidance / outlook (FY3/27, issued 8 May 2026):

FY3/27 company forecast	Value	vs FY3/26
Net sales	¥157.0bn	+5.9%
Ordinary (recurring) profit	¥2.0bn	–58.4%
Net income (owners)	¥3.5bn	+42.2%
EPS	¥64.56	—
Dividend per share	¥20	+¥6

- **Read the guidance carefully — this is the single most important nuance in the print.** FY3/27 net income (¥3.5bn) is **guided above ordinary profit (¥2.0bn)** — the reverse of a normal year. That gap implies **~¥1.5bn+ of non-operating/special gains**, most plausibly the **Osaka land-sale gains (Hirakata land ~¥2.36bn est. gain, delivery May-2026; Mitsuboshi Kaiun parking land ~¥1.30bn est. gain) helping fund the EAF**, and/or accident-related insurance recoveries. **Core operating profitability is still guided sharply lower** (ordinary –58%); the sales "recovery" (+5.9%) mostly reflects a **full year of EAF operation vs the 3-month halt**, not a demand rebound.
- **The dividend hike is real but partly one-off-funded.** ¥20 DPS on ¥64.56 EPS ≈ **31% payout** (consistent with the ≥30% policy) — but that EPS is inflated by the special gains. **Ex-special items, dividend cover on core earnings is thin** — a key watch-item (see forensic/short-seller sections). Still, raising the dividend into a trough is a genuine **shareholder-return / low-PBR-reform** signal.
- **Balance-sheet / cash-flow flags:** the group runs **net cash (~¥6.3bn; ¥9.06bn debt vs ¥15.33bn cash; 72% equity ratio)** — a fortress relative to peers, and a direct target of the retail low-PBR proposals. The forward watch-item is the **~¥95–105bn JV EAF capex through ~2030**, which will consume cash and likely lift gross debt (detail in Section 12).

Market reaction / what's priced in: the stock printed its **2026 low of ¥590 on 8 May 2026 (earnings/guidance day)** — a negative reaction to the weak print and –58% ordinary-profit guide — then **rebounded to ¥680 by 19 May** (dividend hike), settling ~¥603 by end-June 2026. It trades at **PBR ~0.30x** (BPS ~¥2,000), PER ~9.3x (FY3/27e), dividend yield ~3.3% on the ¥20 DPS. At ~0.3x book with net cash the market is pricing **chronic sub-cost-of-capital returns (ROE ~2–5%) and/or capex dilution** — i.e., disbelief in JV value creation and fear of a prolonged spread trough. Re-rating triggers: an ROE inflection, capital-return escalation, JV milestones, or a Nippon Steel move on the (now non-owned) equity.

7. Earnings-Call Summary & Sentiment Shift

What management is focused on (recent briefings, FY3/25 → FY3/26): 1. **The Nippon Steel EAF JV** — the centrepiece: capacity ~2x, low-CO₂ slab, offtake, ~2030 start; framed as the vehicle of "Long-Term Vision 2030." 2. **Stabilising existing EAF operations** — restoring a reliable ~50,000 t/month run-rate after the substation accident; reliability and cost control. 3. **Cost reduction & spread defence** amid imports and weak demand. 4. **Decarbonisation** — 46% supply-chain CO₂ cut by FY2030 vs FY2013; GX positioning/subsidies. 5. **Shareholder returns / capital efficiency** — dividend increases, responding to TSE low-PBR reform. 6. **Real-estate value** — leasing income and latent Osaka Bay land value.

Sentiment arc: **2023 (100th anniversary) constructive** → **FY3/25 knocked back by the substation accident (defensive, apologetic, "recovery")** → **FY3/25–26 cautious/defensive on demand & imports, but structurally more optimistic on the transformational JV and green-steel narrative.** Net: near-term tone is **guarded** (four years of falling profit, ¥2.0bn guide), medium-term tone is **strategically upbeat** on the Nippon Steel partnership. The dividend increase into the trough signals management confidence in downside asset support.

8. Strategic Initiatives (Last ~10 Years) — Status Scorecard

Initiative	Period	Description	Status / Verdict
Blast-furnace/converter shutdown → all-EAF	2002 (legacy)	Exited integrated BF steelmaking after financial distress; shifted to EAF + purchased slab + rolling	Done; structural. Lowered CO ₂ & fixed cost but left a slab-supply dependency the JV now addresses
REVIC-backed rehabilitation	2012–2013	Private restructuring → REVIC (state Regional Economy Vitalization Corp.) support (Mar-2013); banks forgave ~¥60.2bn debt (Aug-2013); ~¥9bn third-party share allotment to 6 sponsors (Nippon Steel & Sumitomo Metal, Hanwa, Nittetsu Shoji/now Nippon Steel Trading, Air Water, Osaka Gas, Daiwa PI Partners) taking >2/3 of votes; resolved negative equity	Success — averted insolvency, reset the balance sheet; established the sponsor relationships (incl. Nippon Steel & Hanwa) that underpin today's alliance
Nippon Steel share exit	Dec 2022	Nippon Steel sold its entire Nakayama stake ("asset compression"); relationship became purely contractual	Neutral for Nakayama; foreshadowed the shift to a JV (rather than parent) structure
2030 Vision / Long-Term Vision	2022–	Long-term vision + sustainability & growth-strategy framework (centenary era)	In progress

Initiative	Period	Description	Status / Verdict
NAKAYAMA STEEL WORKS, LTD. (3408 JP) Decarbonisation roadmap	2021–	Equity Research Deep Dive 46% supply-chain CO ₂ cut by FY2030 vs FY2013; EAF iron-source ratio up; green-steel product	31 Jul 2026 – informational only — not investment advice On track (targets set); execution KPI to watch
Substation accident recovery	FY3/26 (Sep–Dec 2025)	No. 5 substation failure (26 Sep 2025) halted EAF ~3 months (restart 24 Dec 2025; ~¥1.6bn hit); reliability hardening	Recovered; a reliability black mark that triggered the FY3/26 guidance cut
Nippon Steel EAF JV (new large EAF)	2025–~2030	51/49 JV holding co NN Steel LLC (established 1 Apr 2026) ; ~1.2Mt low-CO ₂ EAF at Funamachi, ~¥95–105bn total investment (NS ~¥50bn), NS slab/coil offtake	Signature initiative; execution pending — the whole thesis
3rd Mid-Term Plan (FY2022–24) → new "3-stage-rocket" long-term plan (2025–33)	2022–2033	3rd plan targeted ordinary ¥10bn/ROE 7%/payout ≥30% — met (FY3/24 ordinary ¥12.2bn) before the downturn; new long-term plan (16 May 2025): Phase 1 FY25–27, Phase 2 FY28–30, Phase 3 FY31–33, anchored on the new EAF	3rd plan: succeeded; new plan: early innings
Kanto/Tohoku distribution base (Hitachinaka, Ibaraki)	Dec 2025	New relay-distribution base to widen the domestic sales footprint	Newly opened; commercial ramp
Shareholder-return step-up	2024–2026	Dividend increases (to ¥20 guided FY3/27) amid TSE low-PBR reform	In progress; watch sustainability vs capex

9. Timeline of Key Past Events & Upcoming Catalyst Calendar

Past events (selected): - **1919** — Founder Nakayama Etsuji starts the galvanised-sheet business (Amagasaki); **incorporated 1923**, name 中山製鋼所 from 1934. Historically a fully integrated blast-furnace maker. - **Jul–Sep 2002** — Blast furnace, sinter & converter halted; pivot to EAF (+ proprietary NSR melting). - **2010** — NSR halted → **crude steel becomes EAF-only**. - **2012–2013** — Thick-plate mill suspended; **REVIC-backed rehabilitation (Mar–Aug 2013): ~¥60.2bn bank debt forgiveness + ~¥9bn allotment to 6 sponsors** (incl. NSSMC, which rose to ~20% — Nakayama became an NSSMC equity-method affiliate). - **Jun 2022** — 2030 Vision + 3rd Mid-Term Plan launched. - **Dec 2022** — **Nippon Steel sold its entire ~8.7% stake** (¥799/share, ~¥4.2bn) → 0%; Nakayama becomes independent. - **9 May 2025** — Basic agreement/contract with Nippon Steel on the EAF-holding JV & alliance (up to ¥105.5bn). - **16 May 2025** — New long-term "3-stage-rocket" plan announced. - **26 Jun 2025** — Leadership change: **Naito Nobuhiko → President**; Hakomori Kazuaki → Chairman. - **26 Sep 2025** — **No. 5 substation accident halts the EAF** (~3 months). - **31 Oct 2025** — FY3/26 guidance cut (net income ¥4.2bn → ¥2.3bn). - **6 Nov 2025** — H1 FY3/26 results. - **24 Dec 2025** — **EAF operation restarted** (~¥1.6bn total impact). - **1 Apr 2026** — **NN Steel LLC (EAF JV) established** (Nakayama 51 / Nippon Steel 49). - **8 May 2026** — FY3/26 results (sales ¥148.3bn –12.4%; ordinary ¥4.806bn –40.8%; net income ¥2.462bn –56.8%); FY3/27 guide (ordinary ¥2.0bn; net income ¥3.5bn; DPS ¥20). - **25 Jun 2026** — 132nd AGM; Chairman Hakomori retires to Executive Advisor.

Upcoming catalysts:

Timing	Catalyst	Why it matters
~Aug 2026	Q1 FY3/27 results	First read on FY3/27 trough guide vs. reality
~Nov 2026	H1 FY3/27 results	Spread/import trend; dividend confirmation
FY2026–27	JV EAF detailed engineering / groundbreaking milestones, GX subsidy confirmation	De-risks the ¥95–105bn build; capex phasing
Ongoing	Steel spread / Chinese export & scrap price trends	Direct margin driver
Ongoing	TSE low-PBR reform disclosures, capital-allocation policy updates	Re-rating catalyst at 0.3x book
Medium term	Osaka land sales (Hirakata; Mitsuboshi Kaiun parking) funding the EAF; any further NAV crystallisation	Special gains (drive FY3/27 net income); latent asset value
~2030	New EAF (NN Steel LLC) start-up & ramp	The thesis payoff / risk crystallisation
Watch	Any Nippon Steel move to consolidate (via the JV / a fresh stake)	Potential take-out catalyst (see Section 14) — note NS currently owns 0%

10. Segment & Divisional Breakdown, Geography & Operational KPIs

Reportable segments (3): Iron & Steel (鉄鋼), Engineering (エンジニアリング), Real Estate (不動産). No Chemicals segment (a former Nankai Chemical affiliate was reorganised out in 2025).

Segment revenue (indicative; group is ~98% steel):

Segment	~Sales share	FY3/25–26 direction	Character / margin
Iron & Steel	~97–98%	Down (price/volume + accident)	Cyclical; the swing factor for group P&L. Products: HR coil/strip, thick & medium plate, checkered plate, bars, wire rod, rebar, bar-in-coil, slab, by-products
Engineering	~1%	Down (lower marine-dept. volume)	"Chuki" (rolls, valves, castings) + "Marine" (steel fish/aquaculture reefs, ocean structures). Lumpy, public-works-linked

Segment	~Sales share	FY3/25-26 direction	Character / margin
NAKAYAMA STEEL WORKS, LTD. (3408 JP) Real Estate	<1%	Stable	Equity Research Deep Dive — 31 Jul 2026 — informational only — not investment advice Highest-margin of the three. H1 FY3/26 ~¥708m revenue / ~¥346m ordinary profit (~49% margin). Osaka Bay land via subsidiary Nakayama Kosan

- **Segment operating profit by segment is not granularly disclosed** in accessible sources (it sits in the 決算短信/決算説明会資料 PDFs). Directionally: **Real Estate is small but a high-margin cash cushion; Engineering is small and volatile; Iron & Steel drives ~all of the earnings swing.** The mix has not materially shifted over three years — the group profit decline is essentially the steel spread.
- **Geography:** not separately disclosed; Nakayama is a **predominantly domestic-Japan** producer (construction/civil/machinery/shipbuilding), a **net domestic seller exposed to imports** rather than an exporter. Some Asian export flows via Hanwa's coil-centre network. Export ratio time-series **not obtained**.

Operational KPIs & trends:

KPI	Latest read	Trend / comment
Existing EAF capacity	~600 kt/yr (target ~50,000 t/month)	Reliability target missed in FY2023 (equipment) and FY3/26 (substation accident) ; restoring stable 50kt/month is a core mid-term-plan KPI
New EAF capacity (JV)	~1,200 kt/yr (~2× current)	Under NN Steel LLC; start-up ~2030
Crude-steel self-sufficiency	~50% self-produced / ~50% purchased (of purchased, ~40pts BF slab, ~10pts external EAF)	JV designed to internalise slab (from Nippon Steel, Kobe Steel, Chubu Steel Plate) → structural margin/CO ₂ improvement
Capex intensity	Elevated (JV ramp)	~¥95–105bn JV programme through ~2030 (mostly in the JV) — a step-change from a historically light-capex re-roller
Employees	~780 (parent); avg age 43.8, avg pay ~¥6.5–7.0m	Consolidated headcount not obtained
Order backlog / utilisation / shipment tonnage	Not disclosed / not obtained	In IR decks only
CO ₂ target	~46% supply-chain vs FY2013 by FY2030; CN 2050	~80% of footprint is Scope 3 (purchased iron) → JV reduces it

11. Historical Financials — Income Statement, Per-Share & Returns (Excel-liftable)

Income statement history (¥bn; FY ends 31 March). Firm anchors bolded; earlier years part-estimated/derived — verify vs 有報.

FY (Mar)	Net sales	YoY %	Operating income	Ordinary profit	Net income	Op. margin
FY3/20	136.2	~11% (e)	~1.5 (e)	~1.4 (e)	~1.0 (e)	~1.1%
FY3/21	111.0	~18.6%	~3.8 (e)	~3.6 (e)	~1.8 (e)	~3.4%
FY3/22	166.7	+50.2%	~6.65 (d)	~6.6 (d)	~4.8 (d)	~4.0%
FY3/23	188.5	+13.1%	13.37	13.3	10.2	7.1% ← peak
FY3/24	184.4	~2.2%	12.24	~12.2	8.90	6.6%
FY3/25	169.3	~8.2%	8.12	8.1	~5.69	4.8%
FY3/26	148.3	~12.4%	~4.2	4.806	2.462	~2.8%
FY3/27 (f)	157.0	+5.9%	n/d	2.0	3.5	—

(e)=estimate, (d)=derived from reported growth rates, (f)=company forecast, n/d=not disclosed. EBITDA not cleanly obtainable (D&A not disclosed in accessible sources); one source implied ~¥8bn TTM EBITDA.

Trend read: revenue and profit **peaked in FY3/23** (post-COVID steel super-cycle: sales ¥188.5bn, 7.1% op margin, NI ¥10.2bn), then **four consecutive years of decline** as spreads normalised, demand softened, imports rose, and the FY3/26 substation accident hit. Operating margin compressed from **7.1% → ~2.8%** in three years — textbook spread operating leverage on a ~12% revenue fall.

Per-share & returns (¥ / %):

FY (Mar)	EPS	BPS	DPS	Payout	ROE	ROA
FY3/23	~188 (e)	—	55 (33+22)	~29%	high	—
FY3/24	164.43	—	~55 (e)	~33%	~10%+ (e)	—
FY3/25	105.13	—	~33 (e)	~31%	~5.4%	—
FY3/26	45.42	2,013.25	~14	~31%	2.28%	1.63%
FY3/27 (f)	64.56	—	20	~31%	3.21% (f)	~2.3% (f)

- **Dividend is a ~30–31% payout policy, not a progressive dividend.** DPS fell with earnings from **¥55 (FY3/23 peak) → ~¥14 (FY3/26)**; the FY3/27 "increase to ¥20" is the payout ratio applied to a **special-gain-inflated EPS (¥64.56)** — i.e., mechanical, and a *recovery off a cut* rather than an all-time-high raise. This tempers the "shareholder-return signal" read and is precisely why retail proposals pushed for a **DOE-linked** dividend and buybacks instead.
- **Returns are structurally low:** ROE ~2–3% currently (vs a ~7% mid-term target and ~10%+ in the FY3/23–24 peak), depressed by both trough earnings *and* a very heavy equity base (71.6% equity ratio). ROE ~2–3% at PBR ~0.3x is the core low-capital-efficiency, deep-value profile.
- **Shares:** 63,079,256 issued; **~54.15m ex-treasury** (~8.9m / ~14% treasury) used for EPS/BPS — the treasury block itself is latent buyback/return capacity.

Item (¥bn)	H1 FY3/26 (30 Sep 25)	FY3/26 (31 Mar 26)
Total assets	149.5	~152.4
Net assets (equity)	107.6	~109.1
Equity ratio	72.0%	71.6%
Interest-bearing debt (short+long)	~9.06	~8.5
Cash & equivalents	~15.33	n/d
Net position	Net cash ~¥6.3bn	Net cash
BPS (¥)	—	2,013.25

- **Fortress, net-cash balance sheet** — the defining financial feature. ~¥8.5–9.1bn interest-bearing debt against ~¥108bn equity and ~¥15bn cash → **net cash ~¥6bn; net debt/EBITDA ≈ –0.3x**. Mid-term-plan target net D/E ~0.1x.
- **A dramatic turnaround from 2013**, when the group carried ~¥100bn of interest-bearing debt and required ~¥60bn of bank debt forgiveness (REVIC). Debt has been on a steady, low trajectory since.
- **Credit rating: JCR reportedly BBB, outlook positive** (*confirm grade/date on JCR's page*). No R&I rating identified. Interest costs, current ratio, covenants and undrawn facilities not disclosed in accessible sources.
- **Forward watch-item:** the **~¥95–105bn JV EAF** (mostly inside NN Steel LLC) will consume cash and likely lift gross debt across FY2026–2030 — the net-cash cushion is the war-chest for it, but the balance sheet will lever up from today's ultra-conservative base. FY3/26 operating cash flow ~¥15.3bn (est) gives internal funding capacity; capex/D&A series not obtained.

13. Valuation & Peer Comparison

Nakayama snapshot (as-of late July 2026):

Metric	Value	Note
Share price	¥634 (range ¥590–680 in 2026)	¥603 end-June; ¥590 low on 8-May earnings day
Shares (ex-treasury)	~54.15m	63.08m issued
Market cap	~¥34–38bn	¥34.4bn ex-treasury basis; up to ~¥38–41bn on issued shares
Enterprise value	~¥32.7bn	slight net cash
P/B (PBR)	~0.31x	BPS ¥2,013
P/E (trailing)	~14x	on EPS ¥45.4 (trough)
P/E (forward)	~8–10x	FY3/27e EPS ¥64.56
EV/EBITDA	~4.1x	EBITDA ~¥8bn TTM
EV/Sales	~0.22x	thin-margin
Net debt/EBITDA	~–0.3x (net cash)	—
Dividend yield	~2.2% (¥14) → ~3.3% (¥20e)	—
EBITDA margin	~5.4%	below peer avg
EBIT margin	~3.0%	below peer avg
Sales growth (FY3/26)	~12.4%	trough

Own-history context: since ~2010 Nakayama's PBR has ranged roughly **"loss-making to ~1.3x,"** **chronically below 1.0x**; at ~0.31x it sits near the **bottom of its own decade range**. EV/EBITDA ~4x is also at the low end. On P/B and EV/EBITDA it screens **cheap even versus its own perennially-cheap history** — the new variable absent from history being the Nippon Steel JV re-rating catalyst.

Peer comparison (¥bn; multiples late-July 2026; FY3/26 is a trough year across the group — read forward multiples and P/B, not trailing P/E):

Company	Ticker	Exch.	Type	Mkt cap (¥bn)	P/B	P/E (ttm)	EV/EBITDA	EV/Sales	ND/EBITDA	EBITDA mgn	EBIT mgn	Sales growth
Nakayama Steel	5408	TSE	EAF	~34	0.31	14.0	4.1	0.22	–0.3 (net cash)	5.4%	~3.0%	~12.4%
Nippon Steel	5401	TSE	Integrated	~3,000	0.53	13.3 fwd ¹	~7–8 (e)	~0.9 (e)	~4–5 (e)	~9% (e)	2.4%	+15.7%
JFE Holdings	5411	TSE	Integrated	~1,030	0.40	13.3	5.98	~0.6 (e)	~2–3 (e)	8.1%	~3–4% (e)	~flat
Kobe Steel	5406	TSE	Integrated	~764	0.70	6.2	5.30	~0.6 (e)	~2.1 (e)	11.0%	~7% (e)	~flat
Tokyo Steel Mfg	5423	TSE	EAF (best peer)	~190	0.84	~15.5	~5–6 (e)	~0.35 (e)	net cash	5.7%	2.7%	~18.0%
Osaka Steel	5449	TSE	EAF (NSC ~64% sub)	~83	0.71	NM ²	9.1	0.82	~1.1	9.0%	4.5%	~–8% (e)
Godo Steel	5410	TSE	EAF	~85	0.60	~6	~4–5 (e)	~0.45 (e)	~0.5–1 (e)	7.9%	~5% (e)	~6.5%
Kyoei Steel	5440	TSE	EAF (rebar)	~87	0.39	8.4 fwd	4.47	~0.45 (e)	~1.0 (e)	9.7%	5.4%	~2.4%
Yamato Kogyo	5444	TSE	EAF (H-beam)	~832	~1.0–1.1	~13–14	~14 ³	~1.9 (e)	net cash	13.7%	2.8%	~4.7%

Company	Ticker	Exch	Type	Mkt cap (¥bn)	P/B	P/E (ttm)	EV/EBITDA	EV/Sales	ND/EBITDA	EBITDA mgn	EBIT mgn	Sales growth
Topy Industries	7231	TSE	EAF + wheels	~58–80	0.54	~9–12	4.83	~0.35 (e)	modest	~10% (e)	~5% (e)	~flat
Nucor (context)	NUE	NYSE	EAF (US)	~\$60bn	2.64	15.9 fwd	~11	1.77	~0.9	15.7%	11.6%	+17.2%
Steel Dynamics (context)	STLD	Nasdaq	EAF (US)	~\$39bn	2.36	22.9	15.4	~2.3 (e)	~0.5 (e)	15.1%	~12% (e)	strong

¹ NSC trailing P/E ~175x is meaningless (US Steel one-offs cut NI ~95%); use fwd ~13x. ² Osaka Steel trailing P/E NM on trough earnings. ³ Yamato Kogyo distorted: most profit is equity-method overseas JVs outside consolidated EBITDA — read P/E, not EV/EBITDA. (e)=estimate.

Where Nakayama screens: - **Cheapest name in the peer set on P/B (0.31x)** — below Kyoei (0.39), JFE (0.40), Nippon Steel (0.53), Topy (0.54), Godo (0.60), Kobe (0.70), Osaka (0.71), Tokyo Steel (0.84), Yamato (~1.0). The discount is amplified by the **72% equity ratio, net cash, and latent Osaka real estate** (¥108bn book behind a ~¥34bn cap). - **Cheap but not cheapest on EV/EBITDA (~4.1x)** — in line with the cheap EAF cluster (Kyoei 4.5x, Topy 4.8x, Godo ~4–5x), below Tokyo Steel/Kobe/JFE/Osaka. - **Below-average margins (5.4% EBITDA / ~3% EBIT)** — the fundamental reason for the persistent P/B discount, alongside small scale and a trading-house/strategic-holder-dominated register. - **Verdict:** on asset value **cheap/very cheap**; on earnings power **fair-to-cheap on a normalised basis, optically expensive on trough trailing EPS**. It is a **P/B and asset-backing story, not an earnings-multiple story** — the re-rating requires an ROE inflection, capital-return escalation, JV proof, or consolidation. - **Closest direct comps: Tokyo Steel (5423)** — cleanest listed EAF benchmark; **Osaka Steel (5449) & Godo (5410)** — tightest structural comps (Kansai-region EAF, Nippon Steel orbit); **Kyoei (5440)** for rebar.

14. M&A — Company History, Industry & Takeout Analysis

Nakayama's own M&A (last ~10 years): no material outbound acquisitions — internal reorganisation + one landmark JV.

Date	Action	Nature
2015–2021	Share-exchange consolidations bringing Nakayama Tsusho, Sanko Shoji, Mitsuboshi Kaiun, Sansen Shear, Nakayama Sansei Kenzai to wholly-owned	Group simplification
Dec 2021	Absorption-merger of Nakayama Sansei Kenzai (building materials)	Reorg
Apr 2025	Transferred Nankai Chemical shares to Nankai Global Chemi	Chemical-affiliate divestiture (est.)
1 Apr 2026	Established NN Steel LLC (51 Nakayama / 49 Nippon Steel)	The landmark transaction — the EAF JV

No transaction multiples apply (reorgs/JV, not priced acquisitions). The company has been a consolidator of its own affiliates, not an external acquirer — capital-allocation-conservative.

Industry M&A thoughts. Japanese steel is **structurally consolidating** under Nippon Steel's leadership (Nisshin Steel take-in; Osaka Steel/Godo/Kyoei in the NSC orbit; the ¥/\$ U.S. Steel deal). The **EAF long tail (~30–35 makers, many private regional bar/section producers)** is the fuel for further consolidation as decarbonisation forces scale and capex. Nakayama — sub-scale, deep-below-book, net-cash, and **already 49%-JV'd with Nippon Steel** — is an **unusually clean eventual consolidation candidate**.

Precedent transactions (Japan steel / EAF):

Deal	Year	Structure	Terms / multiple
Nippon Steel + Sumitomo Metal	2012	Merger of equals → NSSMC	Stock, ~no cash premium
Nippon Steel → Nisshin Steel	2017	Cash TOB to ~51%	¥1,620/share tender
Nippon Steel → Nisshin Steel (squeeze-out)	2019 – 2020	Share exchange to 100% + merger	Full take-in of EAF/stainless
Nippon Steel → U.S. Steel	announced 2023 → closed 18 Jun 2025	Cash acquisition	\$55/share, ~40% premium; equity ~\$14.2bn, EV ~\$14.9bn (~7x EV/EBITDA)
Osaka Steel (NSC ~64% sub)	ongoing	NSC-group EAF vehicle	—

Plausible takeout for Nakayama. Japanese steel control deals have generally cleared **~0.8–1.2x P/B, ~5–7x EV/EBITDA, with ~30–40% tender premiums**. From today's **0.31x book**: - **Book-anchored (0.7–1.0x book):** ~¥75–108bn equity ⇒ **~¥1,400–2,000/share (~2.2–3.2x the current ~¥634)**. - **EV/EBITDA (6–7x on ~¥8bn + net cash):** ~¥52–60bn equity ⇒ **~¥960–1,100/share (~+50–75%)**. - The **asset backing (¥108bn book, net cash, Osaka land)** biases a strategic buyer toward the **book-anchored upper end**. A control premium to ~0.8–1.0x book (≈**¥1,500–2,000/share, ~2.5–3x today**) is defensible on precedent — **though no offer is announced**, and Nippon Steel currently owns **0%** of Nakayama (the angle rests entirely on the JV/alliance, not an existing stake).

15. Management Assessment — CEO, Chairman & Key Executives

Archetype: professional-manager-run, NOT founder-family-controlled. Founder Nakayama Etsuji started the business in 1919; after the 2013 dilution/restructuring the **founding family no longer controls or appears among major holders**, and the company is run by career insiders. Aggregate insider ownership is **~2%** — low "skin in the game," typical of post-restructuring Japanese industrials.

Executive	Role	Since	Background / track record	Skin in game
NAKAYAMA STEEL WORKS, LTD. (3408 JP) Nobuhiko Naito (内藤伸彦)	Representative Director & President / CEO	26 Jun 2025	Equity Research Deep-Dive – 31 Jul 2026 Age ~66. Kwansei Gakuin Univ. (Sociology, 1982) → joined Nakayama; Executive Officer 2013, Senior Managing Director 2022, President 2025. Lifelong company insider . Credited with driving the in-house steel-source (自社鉄源) integrated-EAF strategy now embodied in the Nippon Steel JV.	Informational only — not investment advice Minimal (part of ~2% insider block)
Kazuaki Hakomori (箱守一昭)	Rep. Director & Chairman (2025–26)	Chairman Jun-2025	Age ~72. Former President; retired to Executive Advisor (顧問) at the 132nd AGM, 25 Jun 2026.	Minimal
Sachihiro Nakamura (中村佐知大)	Senior Managing Director	—	Executive line	—
Masahiro Morikawa (森川昌浩)	Director & Sr. Managing Exec. Officer	Jan-2026 remit	Oversees General Mgmt, Steelmaking, Rolling, Building-Materials HQ	—
Koji Sumino (角野康治)	Managing Director	—	Executive line	—
CFO	—	—	No separately-titled CFO is publicly designated; finance is handled within the executive line (verify in 有報).	—

Assessment vs. the 5-point framework: 1. **Track record:** Naito is credited with delivering the long-sought self-sufficient EAF steel-source model — the JV is the capstone. But he inherits **four straight years of declining profit**; the turnaround thesis is prospective, not yet proven under his tenure. 2. **Tenure & ownership:** New CEO (Jun-2025); very low insider ownership (~2%) — alignment rests on governance/comp design, not equity stakes. 3. **Capital allocation:** ROE ~5.4% and PBR ~0.31 flag **chronic sub-cost-of-capital returns and over-capitalisation** (equity ratio ~71.6%). The forward test is JV ROIC discipline and the pace of capital return; the dividend step-up (to ¥20) is an early positive signal. 4. **Red flags:** Main governance issue is **related-party concentration** (largest shareholder Hanwa is also a key distributor; Nippon Steel is JV partner/supplier/offtaker) rather than promotional behaviour or excessive comp (director cap ¥300m/yr). 5. **Founder vs. professional:** Fully professional management — appropriate for a restructuring-stage, partner-dependent capital project, but it also means **no controlling owner to force value crystallisation** (raising the importance of the low-PBR-reform/activist and Nippon Steel-consolidation angles).

Recent senior-management changes: June-2025 CEO handover (Naito ← Hakomori); June-2026 Hakomori retirement to advisor; addition of a female outside director (diversity 0 → 8.3%); Morikawa’s expanded production remit (Jan-2026). Company frames FY2026 as **"board continuity."**

16. Board of Directors

Structure: Company with an Audit & Supervisory Committee (監査等委員会設置会社). At the 131st AGM (25 Jun 2025): ~12 officers, 4 outside directors (~33%), 1 female (8.3%). Line-by-line FY2026 continuity "largely maintained" per company disclosure (individual seat confirmation via the 132nd AGM notice was not directly retrievable — treat as indicative).

Director	Role	Independent/Outside	Background / affiliations
Kazuaki Hakomori (箱守一昭)	Rep. Director & Chairman	Executive	Ex-President; retired to Executive Advisor 25 Jun 2026
Nobuhiko Naito (内藤伸彦)	Rep. Director & President/CEO	Executive	Career insider since 1982; CEO since 2025
Sachihiro Nakamura (中村佐知大)	Senior Managing Director	Executive	Executive line
Masahiro Morikawa (森川昌浩)	Director, Sr. Managing Exec. Officer	Executive	Production/manufacturing HQ (from Jan-2026)
Koji Sumino (角野康治)	Managing Director	Executive	Executive line
Masahiro Nakatsukasa (中務正裕)	Director	Outside	Attorney (Osaka legal background)
Sayuri Murakami (村上早百合)	Director	Outside	Female outside director (unrelated to Murakami/City Index activist funds)
Ryohei Kishida (岸田良平)	Director, Audit & Supervisory Cmte (full-time)	Internal (standing)	Standing audit-committee member
Masaya Tsunoda (角田昌也)	Director, Audit & Supervisory Cmte	Outside	Outside audit-committee member
Kazuyoshi Tsuda (津田和義)	Director, Audit & Supervisory Cmte	Outside	Outside audit-committee member

Board changes, last ~5 years: 2013 restructuring reshaped the board around creditor/sponsor supporters; recent moves are the **2025 CEO transition, 2026 chairman retirement, a new female outside director** (diversity), and Morikawa’s expanded remit. **No Nippon Steel-nominated director is currently identifiable** (consistent with Nippon Steel’s 2022 share exit).

17. Top Shareholders & Activism

Top shareholders (~FY3/25 register; holders 4–7 are best-available/partially unverified — reconcile with the 有価証券報告書):

#	Shareholder	%	Nature
1	Hanwa Co. (阪和興業, 8078)	14.87%	Steel-trading house; 2013 rescue investor; also a key distributor/off-taker (related party)
2	Master Trust Bank of Japan (trust)	10.02%	Nominee/trust custodian (institutional beneficiaries)

#	Shareholder	%	Nature
NAKAYAMA STEEL WORKS, LTD. (3408 JP)			Equity Research Deep-Dive
3	Air Water Inc. (エア・ウォーター)	8.73%	01 Jul 2026 – informational only — not investment advice Industrial-gas group; 2013 sponsor; cross-holding
4	Maruichi Steel Tube (丸一鋼管)	~4.9% (e)	Steel-pipe maker; cross-holding
5	Osaka Gas (大阪ガス)	~3.5% (e)	2013 sponsor; cross-holding
6	Amagasaki Seikan (尼崎製鐵, entity unverified)	~2.4% (e)	Cross-holding
7	Nippon Steel Trading (日鉄物産)	~1.5% (e)	Nippon Steel-group trading co; 2013 sponsor; residual "NS-family" holder
—	Custody Bank of Japan; Mizuho/SMBC-side banks; employee shareholding assoc. (中山持株共栄会)	n/a (e)	Trust nominees / lenders / employees

- **Shares issued: 63,079,256** (May 2025); **treasury stock not obtained (unverified)** — a material treasury block is plausible given the over-capitalised balance sheet (see valuation note on share-count/market-cap range).
- **Nippon Steel = 0%**. Entered as a 2013 sponsor (peaked ~20%, making Nakayama an equity-method affiliate of then-NSSMC), then **sold its entire ~8.7% stake in Dec-2022 at ¥799/share (~¥4.2bn) to Nomura** as part of its strategic-shareholding-reduction/decarbonisation programme. Nakayama is now **independent** — not consolidated or equity-affiliated with Nippon Steel.
- **Ownership structure:** individual investors ~45%; strategic/cross-holding corporates ~35%+ ("friendly" block). High headline float but a large friendly block = **structural defence against activism**.

Activist / shareholder campaigns: - **131st AGM (26 Jun 2025): two individual shareholders** (not a brand-name fund) filed proposals; board **opposed all** (resolved 23 May 2025). Outcomes (all rejected): **dividend increase to a 3.5% DOE** — ~**13.1%** support; **share buyback(s)** — ~**11.8%** and ~**13.7%** support. Effectively a **retail-led low-PBR/return-of-capital campaign**. - **No dedicated activist-fund campaign** (Murakami/City Index, Strategic Capital, 3D, etc.) against Nakayama was found; Strategic Capital's steel target is **Osaka Steel (5449)**, not Nakayama. The friendly ~35% block + Hanwa's 14.9% make Nakayama a hard activist target despite its cheapness (absence of a fund campaign is "not found," not disproven). - **TSE low-PBR reform** is the live pressure: PBR ~0.31, equity ratio ~71.6%, ROE ~5.4% — a textbook "conscious of cost of capital" target. Company responses: first **Integrated Report (2024)**, mid-term plan, and the **8 May 2026 dividend increase (to ¥20)**.

18. Stakes Held in Other Companies (Subsidiaries, Affiliates, Cross-Holdings)

Consolidated subsidiaries (group; ownership as company-stated): - **Nakayama Tsusho (中山通商)** — group core steel-trading/distribution arm (est. 1948). - **Sanko Shoji (三星商事)** — trading. - **Mitsuboshi Kaiun (三星海運)** — shipping/logistics (its Osaka parking-lot land is being sold — est. gain ~¥1.30bn — to help fund the EAF). - **Sansen Shear (三泉シェアー)** — steel secondary processing/shearing. - **Nakayama Kosan (中山興産)** — real-estate trading, brokerage & services (drives the Real Estate segment).

Equity-method affiliate: - **Nippon Steel Bolten (日鉄ボルテン)** — high-tension bolt maker; **Nakayama 15% / Nippon Steel 85%**. Strategically minor but symbolic of the continued Nippon Steel commercial linkage.

Forthcoming JV (from ~Mar 2026): - **NN Steel LLC (NN製鋼合同会社)** — **Nakayama 51% / Nippon Steel 49%**. Owns and leases the new ~1.2Mt low-CO₂ EAF at Funamachi; the single most strategically and financially relevant related entity (see Sections 5, 8, 14).

Relevance: subsidiaries are operationally integral (distribution, logistics, processing, real estate); the Nippon Steel Bolten stake is immaterial financially; **NN Steel LLC is the defining forward-looking holding** and the fulcrum of the equity story.

19. Adversarial Deep-Dive: Bull Case, Bear Case, Pre-Mortem, Contrarian View

19.1 Bull Case

1. **Deep asset-value floor (0.3x PBR)**. At ~0.3x book with a net-cash-ish/asset-rich balance sheet and latent Osaka Bay real-estate value, downside to tangible/replacement value looks limited; the TSE low-PBR reform gives management a mandate (and cover) to close the gap via returns and disclosure.
2. **Transformational, subsidised, de-risked growth option**. The Nippon Steel 51/49 EAF JV roughly doubles crude-steel capacity to ~1.2Mt, internalises slab (kills the structural slab-purchase drag), delivers a low-CO₂ "green slab" product with a **contracted Nippon Steel offtake**, and is co-funded ~¥50bn by Nippon Steel plus likely GX subsidies. Few ¥33bn-cap companies control a ¥100bn strategic project with a giant partner absorbing risk.
3. **Decarbonisation tailwind**. All-EAF route + new low-CO₂ EAF aligns with carbon pricing/green procurement; potential green-steel premium and subsidy capture.
4. **Consolidation optionality**. Nippon Steel's 49% JV co-control and slab offtake make Nakayama a logical eventual full-consolidation target — a discrete take-out catalyst at a premium to a 0.31x-book price (see Section 14). Note Nippon Steel owns **0%** of Nakayama today, so this is an alliance/JV-driven angle, not an existing-stake creep.
5. **Shareholder-return inflection**. Dividend raised to ¥20 into an earnings trough signals confidence and a policy shift; capital-return upside as low-PBR reform bites.
6. **Earnings trough / operating leverage**. Four years of decline and a ¥2.0bn ordinary-profit guide set a low bar; any spread normalisation (import relief, demand recovery, energy cost easing) drives outsized recovery given spread operating leverage; the substation-accident base effect also flatters comparisons.

19.2 Bear Case (2–3 permanent-impairment risks)

NAKAYAMA STEEL WORKS, LTD. (5408 JP)

Equity Research Deep-Dive · 31 Jul 2026 · Informational only — not investment advice

1. **Structural spread compression from imports + energy.** If Chinese/Asian overcapacity persists and Japanese electricity stays high, an sub-scale EAF re-roller can be **permanently squeezed below its cost of capital** — 0.3x book may be the *correct* price for a value-destructive commodity asset, not a discount. The ¥2.0bn ordinary guide (≈1.4% of sales) is perilously thin.
2. **Capex supercycle destroys the balance sheet / dilutes returns.** A ¥95–105bn EAF build on a ¥33bn-cap company — even with Nippon Steel's 49% and offtake — loads leverage, depreciation and execution risk. If steel markets are weak at ~2030 start-up, or costs overrun, ROIC on the JV could sit **below WACC for years**, converting the "growth option" into a value trap and pressuring the dividend.
3. **Value leakage to the JV partner / channel.** Nippon Steel is JV co-owner (49%) *and* slab supplier *and* slab/coil offtaker, while Hanwa is the largest shareholder *and* a principal distributor. Offtake pricing, JV governance and distribution terms can be structured to favour the counterparties — a **related-party web** that can cap the value Nakayama's own minority shareholders capture from any upside (even though Nippon Steel is not itself a Nakayama shareholder).

19.3 Pre-Mortem (it's 2029 and the equity has halved — what happened?)

- Steel spreads stayed in a secular trough; the existing EAF ran below 50kt/month again (another reliability incident); FY3/27–29 ordinary profit hovered near/below ¥2bn.
- JV EAF capex ballooned past ¥105bn and/or slipped past 2030; interest cost and depreciation swamped a weak steel market at ramp; free cash flow turned deeply negative and the dividend was cut.
- Real-estate value never crystallised; PBR stayed at 0.3x because ROE stayed ~2–4%.
- Nippon Steel used offtake/JV terms to capture the green-slab economics, leaving Nakayama minorities with the capex and little of the margin.

19.4 Contrarian View — What the Market May Be Refusing to See

The consensus reads 5408 as a *melting ice cube* (four years of falling profit, ¥2.0bn guide, imports forever). The contrarian read: the market is **anchoring on the trough P&L and ignoring the balance sheet and the option**. It is pricing the JV as pure capex risk with zero credit for (a) the *elimination of the slab-purchase drag* (a permanent structural margin uplift once the EAF runs), (b) a *contracted anchor customer* (Nippon Steel) that de-risks utilisation, (c) *GX subsidy* capture, and (d) the rising probability that **Nippon Steel simply buys in the minority** rather than run a listed 51%-controlled JV vehicle indefinitely — a scenario in which 0.3x book is a gift. The market is refusing to see that this is less a steel-cycle stock than a **cheap call option on Nippon Steel's Kansai EAF strategy**, with a real-estate-backed strike price.

20. Scenario Analysis (Illustrative) — Bull / Base / Bear

Framework, not a formal target. Anchored on ~54.15m shares (ex-treasury), ~¥34bn market cap, ¥634 reference price, BPS ¥2,013, ~0.31x current PBR, ~¥8bn mid-cycle EBITDA, and net cash. All figures illustrative — rebuild in a live model. Primary valuation lens is P/B (an asset/return story), cross-checked on EV/EBITDA.

Scenario	Key assumptions	Norm. ordinary profit	Exit multiple	Implied equity value	Implied / share	vs ¥634
Bull	Spread recovery + JV delivers structural slab-cost/capacity benefit; consolidation/take-out or genuine ROE inflection toward ~7%+; re-rate to ~0.7–1.0x book	~¥8–10bn	0.7–1.0x book (take-out at/near NAV)	~¥76–109bn	~¥1,410–2,010	+120% to +215%
Base	Spreads range-bound; JV on time/budget; ROE drifts to ~4–5%; modest capital-return step-up; re-rate to ~0.45–0.55x book	~¥5–6bn	0.45–0.55x book	~¥49–60bn	~¥905–1,110	+45% to +75%
Bear	Secular import squeeze; JV capex overruns / weak ~2030 ramp; dividend cut once land gains fade; PBR stuck ~0.28–0.32x	~¥1–2bn	0.28–0.32x book	~¥31–35bn	~¥565–645	–11% to ~flat

Read: the distribution is **positively skewed** — the bear case is roughly "price stays near here / mild downside" (the ~0.3x-book, net-cash, asset-backed floor), while base/bull offer material upside, heavily contingent on (i) spread normalisation, (ii) JV execution, and (iii) a re-rating driver (return policy, ROE inflection, or Nippon Steel consolidation). The **key swing variables are the exit PBR (re-rating) and normalised spread — not revenue growth**. The catch: the bear case can persist as "**dead money**" for years even without permanent impairment, because 0.3x book with ~2–3% ROE is a *stable* equilibrium absent a catalyst.

21. Forensic Accounting Review — Red Flags & Risk Screen

General screen on a Japanese small-cap steelmaker; specific line items to be verified against the 有価証券報告書 (annual securities report) and 決算短信.

Area	What to scrutinise at 5408	Flag level
Revenue recognition	Steel is largely point-in-time on shipment via trading houses — low manipulation risk. Watch trading-house channel (bill-and-hold?, consignment), and Engineering percentage-of-completion on long marine/construction projects (estimate-driven).	Low–Med (Engineering POC)

Area	What to scrutinise at 5408	Flag level
NAKAYAMA STEEL WORKS, LTD. (3408 JP) Segment reporting	Only 3 segments; Real Estate margins & asset values deserve scrutiny (latent gains, intra-group leases). Confirm segment operating-profit reconciliation and any unallocated corporate costs.	Equity Research Deep-Dive - 31 Jul 2026 - Informational only — not investment advice Med (disclosure granularity)
Related-party transactions	Two structural webs: (1) Nippon Steel = JV co-owner (49%) + slab supplier + slab/coil offtaker (note: 0% shareholder since 2022); (2) Hanwa = largest shareholder (14.9%) + principal distributor . Also the 15%-owned Nippon Steel Bolten affiliate. Scrutinise slab purchase/offtake pricing, JV funding terms, and Hanwa trading volumes/margins in the RPT note.	High (structural)
Leases	Real-estate leasing income and any sale-and-leaseback; IFRS/JGAAP lease treatment on plant/land.	Low–Med
Contingencies	Post- substation-accident liabilities/insurance recoveries; environmental/soil remediation on old Osaka Bay industrial land; JV guarantees/commitments (¥100bn programme).	Med–High (env & JV commitments)
Stock-based comp	Typically minimal at Japanese industrials; confirm any executive stock/BIP trust.	Low
Goodwill / intangibles	Limited acquisitive history → low goodwill risk; verify any JV equity-method carrying values and impairment triggers if steel weakens.	Low–Med
Capex / capitalisation	The ¥95–105bn JV: watch capitalised interest, JV equity-method vs. proportional treatment, subsidy accounting (grant offset vs. deferred) , and depreciation policy at ramp — big swing factors for reported profit ~2029–31.	Med–High (going forward)
Inventory	Steel/scrap price swings → lower-of-cost-or-market write-downs; inventory-valuation method (moving-average) can flatter/hurt COGS in volatile scrap markets.	Med (cyclical)
Cash-flow quality	Reconcile ordinary profit to operating cash flow; watch working-capital releases masking weak earnings, and the shift to negative FCF as JV capex funds.	Med

Bottom line: the dominant forensic issue is **not aggressive accounting but structural related-party dependence on Nippon Steel** (pricing of purchased slab, JV terms, offtake), plus **capex/subsidy accounting** on the mega-project and **environmental/accident contingencies**. Earnings quality is otherwise typical of a conservative Japanese industrial (low goodwill, tangible-asset-heavy, cash-backed).

22. Short-Seller Brief — Dismantling the Bull Case

Thesis: 0.3x book is a value trap, not a floor. - **What structurally breaks the model:** a sub-scale EAF re-roller with **no pricing power** on commodity grades, **high Japanese power costs**, and a **permanent Chinese/Asian import overhang** earns below its cost of capital through the cycle. The ¥2.0bn FY3/27 ordinary-profit guide is ~1.4% of sales — one bad scrap/electricity/spread quarter from breakeven. - **Where revenue is concentrated & the risk:** ~90%+ in one **Iron & Steel** segment, one **Osaka site**, largely **one end-market (domestic construction)** intermediated by powerful **trading houses**. A construction downturn, another **single-site accident** (as in FY3/25), or trading-house destocking hits disproportionately. - **Why the moat is weaker than bulls think:** the "moat" is really **dependence on Nippon Steel**. That partner is also a competitor-adjacent giant that can dictate slab and offtake terms; the JV is *Nippon Steel's* Kansai EAF strategy with Nakayama as the funding/host vehicle. Minorities may fund the capex and see the margin routed to the parent. - **Most dangerous competitor bulls underestimate: Tokyo Steel Mfg (5423)** — a far larger, lower-cost, financially stronger EAF operator with its own flat-product ambitions and superior scale economics; and, more broadly, **imported flat steel**. Nakayama has no cost answer to either. - **Worst capital allocation risk:** committing to a **¥95–105bn** EAF at the top of a capex-cost cycle into a structurally oversupplied product — classic **cyclical-capex-into-glut** value destruction; even a 49% Nippon Steel share leaves Nakayama carrying enormous relative exposure. - **Related-party / incentive concerns:** the largest shareholder (Hanwa, 14.9%) is also a key distributor; Nippon Steel is JV co-owner + slab supplier + offtaker; low insider ownership (~2%) means alignment rests on governance, not equity. Scrutinise transfer/offtake pricing. - **What must hold for the price to be justified:** stable-to-improving spreads, on-time/on-budget JV, a real-estate/return re-rating, *and* Nippon Steel structuring the JV/offtake in Nakayama's favour. **The dividend is weaker than it looks:** FY3/27's ¥20 DPS is set by a ~31% payout on an EPS (¥64.56) **inflated by ~¥1.5bn of one-off land-sale gains**; on *core* earnings the run-rate DPS is closer to **~¥8–12**. Once the land gains are spent (post-FY3/27), a payout-ratio policy applied to trough earnings mechanically **cuts the dividend** unless spreads recover — removing the yield support just as JV capex peaks. **If ordinary profit disappoints 20–30% from the already-low ¥2.0bn guide, core dividend cover evaporates.** - **Single permanent-impairment scenario (and plausibility):** JV EAF completes into a weak-2030 steel market with cost overruns; depreciation + interest overwhelm a trough spread; FCF turns deeply negative; dividend cut; PBR stays ~0.25x. **Plausibility: moderate** — the balance sheet and Nippon Steel backing argue against outright insolvency, but a decade of sub-WACC returns is entirely plausible.

Short-seller's honest caveat: the 0.3x-book, asset-backed, Nippon-Steel-anchored setup makes this a **poor outright short** (limited downside, take-out/parent-buy-in tail risk, dividend support). The bear case is better expressed as **"dead money / value trap"** than "structural zero."

23. Fifteen Questions for the CEO (Ordered by Information Value)

- JV economics:** What after-tax **ROIC** do you underwrite for the ¥95–105bn new EAF, at what mid-cycle spread and utilisation, and what is the break-even steel price?
- Nippon Steel offtake terms:** How is the slab offtake **priced** (formula, benchmark, floor), and how do you assure minority shareholders the terms are arm's-length?
- Funding:** How is Nakayama's share of the ¥95–105bn funded (debt/equity/subsidy/cash), what is the **peak net-debt/EBITDA**, and is the dividend protected through the build?

5. **Spread defence:** How much of your book can you **re-price** against scrap/energy, and what is your structural cost gap vs. imports and vs. Tokyo Steel?
6. **Capital allocation / PBR:** At 0.3x book, why build a ¥100bn asset rather than **return capital / buy back stock**? What ROE closes the PBR gap, and by when?
7. **Consolidation:** Given the 51/49 JV, is a **full combination with Nippon Steel** (or a fresh Nippon Steel stake) contemplated, and how would the board protect minority value if approached — especially since Nippon Steel exited the register in 2022?
8. **Green-steel monetisation:** What **green premium** and **GX subsidy** are actually contracted vs. aspirational, and how much JV capex do subsidies offset?
9. **Real estate:** What is the **market/appraised value** of the Osaka Bay land vs. book, and what is the plan (develop, lease, monetise)?
10. **Mid-term plan targets:** What are the explicit FY2027 **ROE/ROIC/margin** targets, and what happens to the plan if spreads don't recover?
11. **Import scenario:** If Chinese flat-steel exports stay elevated for 3–5 years, what is your **downside operating plan** (curtailment, mix shift, cost)?
12. **Engineering & Real Estate:** Are these **core**? Would you divest to fund the EAF or simplify the story?
13. **Energy:** What is your **electricity-cost hedging/procurement** strategy and exposure to Kansai Electric tariff moves; any captive/renewable power plan for the new EAF?
14. **Governance:** How many **truly independent** directors oversee the Nippon Steel-related transactions, and is there a special committee for RPTs?
15. **Succession & incentives:** Post the 2025 CEO transition, how is management **incentivised on ROIC/TSR** (not volume), and what is insider ownership?

24. Sources, Methodology & Key Verification Items

Methodology. Figures were triangulated across public sources (multiple corroborating sources sought for each headline number). The issuer's own domain (nakayama-steel.co.jp) and several data aggregators were not directly retrievable in this environment; where a figure rests on a single source or required derivation it is flagged "(e)/(est)/(d)/(unverified)". **This note is a synthesis for framing and model-building — verify all cells against primary filings before use in a live model or investment decision.**

Primary sources to pull before finalising a model (highest-value, currently to-verify): 1. FY3/26 決算短信 (Tanshin) & 決算説明会資料 (8 May 2026) — exact operating income, **segment operating profit by segment**, cash-flow statement (**capex, D&A, OCF/FCF**), crude-steel tonnage & utilisation. 2. FY3/25 有価証券報告書 (annual securities report) — top-10 shareholder table & exact %s (holders 4–10), **treasury-stock count**, director remuneration, **related-party-transaction note** (Hanwa, Nippon Steel Trading, NN Steel LLC), geographic/export split, lease & contingency notes. 3. **NN Steel LLC JV disclosures** (Nippon Steel PR 9 May & 26 Nov 2025) — exact investment phasing, equity contributions, offtake terms, capacity, start-up date. 4. **31 Oct 2025 downward-revision & 17 Oct 2025 accident disclosures** — substation-accident financial detail. 5. **JCR rating page** — confirm grade/outlook (reported BBB/positive). 6. **New long-term "3-stage" plan (16 May 2025)** & mid-term plan — explicit ROE/ROIC/margin/capex targets.

Representative sources consulted (all public): Nikkei / Nikkei disclosure (TDnet), Kabutan, IR Bank, Minkabu, Yahoo! Finance Japan (Fisco/Wealth Advisor/iFIS), Buffett-Code, MarketScreener, StockAnalysis, Simply Wall St, TipRanks, TradingView, Reuters, Japan Metal Daily / Japan Metal Bulletin, MONOist, Argus, SteelOrbis, REVIC, Nippon Steel newsroom, Global Energy Monitor, Transition Asia, Mizuho/Kinzai industry reports, Wikipedia (ja/en), company IR references. Multiples/prices as-of **late July 2026** and move daily.

Highest-confidence facts: 3-segment structure; 2002 blast-furnace shutdown; 2013 REVIC rehabilitation (6 sponsors); **Nippon Steel's full 2022 share exit → 0%**; the 2025–26 **NN Steel LLC 51/49 JV** (1.2Mt EAF, up to ¥105.5bn, start ~2030); **26 Sep 2025 substation accident → 24 Dec 2025 restart (~¥1.6bn)**; FY3/26 actuals (sales ¥148.3bn / ordinary ¥4.806bn / NI ¥2.462bn); FY3/27 guide (ordinary ¥2.0bn / NI ¥3.5bn / DPS ¥20); **Hanwa 14.87% top shareholder**; PBR ~0.31x; 72% equity ratio / net cash; CEO Naito (Jun-2025); Nakayama's 15% stake in Nippon Steel Bolten.

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